

Technical Document #643: Software Engineering - Comprehensive Overview

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API design defines how software components communicate. RESTful APIs and GraphQL are popular API design approaches.

Refactoring improves code structure without changing behavior. It makes code easier to understand, maintain, and extend.

Technical debt represents the cost of choosing easy solutions over better ones. It accumulates over time and slows development velocity.

Domain-driven design (DDD) models software around business domains. It improves communication between technical and business stakeholders.

Code review examines code changes before merging. It improves code quality, catches bugs, and shares knowledge across the team.

Test-driven development (TDD) writes tests before writing code. It improves code quality and design through small, incremental changes.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Key Takeaways:

- Microservices architecture structures applications as independently deployable services.
- Code review examines code changes before merging.
- Domain-driven design (DDD) models software around business domains.